Week 5, October 1, 2025

Main topics

- Bayesian Inverse Problems
- Research lecture

Material
From last week: Sections 3-3.2.1 (pp. 49-58) and 3.4 (pp 72-79) from the book Statistical and Computational Inverse Problems by Jari Kaipio and Erkki Somersalo. Spend 20 minutes with the CUQI-Book.
Prepare for the research lecture by browsing the CUQI homepage.

Program

13:00-13:30 Inverse Problems and randomness

13:00-15:30 Exercises, see below

15:45-17:00 Research lecture: Per Christian Hansen on CUQI:Computational Uncertainty Quantification for Inverse Problems

Exercises
You are highly encouraged to discuss and solve the exercises in small groups.

1. Go through Probably the simplest BIP in the world (the short story). You can run the Python code in browser if you press the rocket and choose Live Code.
2. Assume that X and E are independent Gaussian random variables with prior and noise models

$$X \sim \mathcal{N}(0, \gamma^2 I),$$
$$E \sim \mathcal{N}(0, \sigma^2 I).$$

Bayes' formula then shows that the posterior is Gaussian with density

$$\pi_{\text{post}}(x) = \pi(x|y_{\text{obs}}) = \frac{\pi(y_{\text{obs}}|x)\pi_{\text{pr}}(x)}{\pi(y)} \propto \exp\left(-\frac{1}{2}(x-\overline{x})^T \Gamma_{\text{post}}^{-1}(x-\overline{x})\right).$$

Here

$$\Gamma_{\text{post}} = \gamma^2 - \gamma^4 A^T(\gamma^2 AA^T + \sigma^2)^{-1}A,$$
$$\overline{x} = \gamma^2 A^T(\gamma^2 AA^T + \sigma^2)^{-1} y_{\text{obs}}.$$

- Show using the SVD of A that $\overline{x} = R_\alpha y_{\text{obs}}$ with $\alpha = \sigma^2/\gamma^2$.

- Interpret that higher noise variance requires stronger regularization, and that lower prior variance requires higher regularization. What is your intuition?
- Show that the posterior (in the sense of positive definite matrices) satisfies $\Gamma_{\text{post}} < \gamma^2$. Interpret, that "measurements decrease the uncertainty."

3. We can simulate a Bayesian inverse problem by starting from a deterministic x^*, calculating $y^* = Ax^*$, and then sample the noise e iid from a Gaussian $e \sim \mathcal{N}(0, \gamma^2 I)$, before computing $x_\alpha = R_\alpha(y^* + e)$.
   Use the Abel problem with $N = 256$, $\nu = 1.5$, and the functions $x(s) = s^\nu$, $y(t) = t^{\nu+1/2}\beta(1+\nu, 1/2)$. Let the noise and prior variance be defined by $\gamma = 0.02$ and $\sigma = 0.005$, respectively, and define $\alpha = \sigma^2/\gamma^2$. Repeat, say 100 times, the experiment of sampling the noise, solving the Tikhonov regularized problem $x_\alpha = R_\alpha(y^* + e)$. Plot the obtained distibution of solutions. Compute the sample mean.

4. Show that in the above scheme with $e \sim \mathcal{N}(0, \gamma^2 I)$, you sample from the distribution $R_\alpha(y^* + e) \sim \mathcal{N}(R_\alpha y^*, \Gamma),\ \Gamma = \sigma^2 R_\alpha R_\alpha^T$. Please help me understand that this is the right porterior, $\Gamma = \Gamma_{\text{post}} = \gamma^2(I - R_\alpha A) \ldots$

5. Repeat the above experiment with different noise models (e.g. uniform noise, other mean values etc). Are you sampling from the posterior distribution?

6. Take a look at chapter 3 in the online CUQIpy book; go back to the previous notebooks as you need.